

Data Availability and Provenance Statements

INSTRUCTIONS: Every README should contain a description of the origin (provenance), location and accessibility (data availability) of the data used in the article. These descriptions are generally referred to as “Data Availability Statements” (DAS). However, in some cases, there is no external data used.

- ☐ This paper does not involve analysis of external data (i.e., no data are used or the only data are generated by the authors via simulation in their code).

Sessions were run using z-tree (Fischbacher, 2007). Each session began with instructions. Participants had printed copies of the payoff tables for all five games. Sessions were run at the LINEEX lab at the University of Valencia, with undergraduate students recruited from the university as participants. The payoffs were denominated in Experimental Currency Units, with 1 ECU = 0.2€. Participants received their cumulative earnings for all rounds. Including a 5€ show-up fee, average pay was 19.90€. Sessions lasted approximately an hour.

Instructions for each treatment are contained in the folder labelled “Instructions.” The z-tree software used for each treatment is in the folder labelled “Z-Tree Programs.” See the ReadMe file in this folder for details about which program was used for each treatment.

Statement about Rights

- ☒ I certify that the author(s) of the manuscript have legitimate access to and permission to use the data used in this manuscript.
- ☒ I certify that the author(s) of the manuscript have documented permission to redistribute/publish the data contained within this replication package. Appropriate permission are documented in the [LICENSE.txt](#) file.

All data was gathered at the LINEEX lab at the University of Valencia and is owned by the authors. As per IRB regulations, the data being shared is *deidentified*.

Summary of Availability

- ☒ All data **are** publicly available.
- ☐ Some data **cannot be made** publicly available.
- ☐ **No data can be made** publicly available.
- ☐ Confidential data used in this paper and not provided as part of the public replication package will be preserved for ___ years after publication, in accordance with journal policies.

Details on each Data Source

All data used in this paper comes from experiments conducted at the LINEEX lab at the University of Valencia. Individuals using this data should cite the EJ paper in which the

experimental results were presented: Brandts, Jordi and David J. Cooper (2025), “Managerial Leadership, Truth-Telling, and Efficient Coordination”, forthcoming *The Economic Journal*.

Dataset list

Data file	Source	Notes	Provided
BC Master Data, Main Experiment.dta	LINEEX	Raw data for all treatments except EC - MC	Yes
BC Coding Data, Main Experiment.dta		Coding results for chat treatment (CH/S – D, CH/A – D, and CH – MC)	Yes
BC Master Data, Follow Up Experiment.dta		Data for EC – MC and Section 5	Yes

Computational requirements

Software Requirements

- ☐ The replication package contains one or more programs to install all dependencies and set up the necessary directory structure.
- All of the do files were run using Stata 18.5.

Controlled Randomness

- ☐ Random seed is set at line ____ of program ____
- ☒ No Pseudo random generator is used in the analysis described here.

Memory, Runtime, Storage Requirements

INSTRUCTIONS: Memory and compute-time requirements may also be relevant or even critical. Some example text follows. It may be useful to break this out by Table/Figure/section of processing. For instance, some estimation routines might run for weeks, but data prep and creating figures might only take a few minutes. You should also describe how much storage is required in addition to the space visible in the typical repository, for instance, because data will be unzipped, data downloaded, or temporary files written.

Summary

Approximate time needed to reproduce the analyses on a standard 2025 desktop machine:

- ☒ <10 minutes
- ☐ 10-60 minutes
- ☐ 1-2 hours
- ☐ 2-8 hours
- ☐ 8-24 hours

- ☐ 1-3 days
- ☐ 3-14 days
- ☐ > 14 days

Approximate storage space needed:

- ☒ < 25 MBytes
- ☐ 25 MB - 250 MB
- ☐ 250 MB - 2 GB
- ☐ 2 GB - 25 GB
- ☐ 25 GB - 250 GB
- ☐ > 250 GB
- ☐ Not feasible to run on a desktop machine, as described below.

Details

- The do file was run on a MacBook Pro with a 2.3 GHz Quad-Core Intel Core i7. The memory was 32 GB 2733 MHz LPDDR4X. MacOS was Sonoma 14.7.1.

Description of programs/code

There is a single Stata do file, “Brandts and Cooper, EJ 2025, replication package, full do file,” which creates all of the variables (beyond the raw data) used in the analysis and generates all of the numbers reported in the paper. The do file is broken into three sections. The first section loads the raw choice data from the main treatments and creates the relevant variables. It then produces the analysis for Sections 4.1 and 4.2.a as well as a small number of figures used elsewhere in the paper. The next section loads all of the coding data, creates relevant variables, and produces the analysis for Section 4.2.b as well as a small amount of analysis used elsewhere in the paper. The final section of the do file loads the raw data for the follow-up treatment (EC – MC) and produces the analysis for Section 5.

User of the do file should note that the default directory is set at the beginning of the do file. This will need to be changed to reflect the location of the data files.

Also note that `outreg2` will open a data editor and stop the do file each time it is run. We have commented out these commands, but if you want to see the tables produced by `outreg2` you will need to remove the comments. If you run `outreg2`, you will need to shut the data editor and hit enter to continue the do file.

Instructions to Replicators

All of the numbers in the paper are generated using the Stata do file “Brandts and Cooper, EJ 2025, replication package, full do file”. Before running the program, you will need to put the three data files “BC Master Data, Main Experiment.dta”, “BC Coding Data, Main Experiment.dta”, and “BC Master Data, Follow Up Experiment.dta” into a directory. All of the variables in the data files have been labeled with descriptions of the variable; this information is replicated below. Be certain that you set the default directory in the second line of the do file. Note that `outreg2` will open a date editor and stop the do file each time it is run. We have commented out these commands, but if you want to see the tables produced by `outreg2` you will need to remove the comments. If you run `outreg2`, you will need to shut the data editor and hit enter to continue the do file.

Running the do file will generate all of the numbers used in the paper, including the appendices. The do file will also generate all of the necessary variables beyond those already in the raw data files. There are extensive comments in the do-file letting you know what numbers are being generated by any particular section of the do file.

Variable List, BC Master Data, Main Experiment

<u>Variable Name</u>	<u>Description</u>
<code>true_treatment_ordered</code>	treatments, same order as paper
<code>session</code>	session id
<code>part_decent</code>	ND - D
<code>part_cent</code>	SC - MC
<code>str_decent</code>	STR - D
<code>str_cent</code>	STR - MC
<code>hsl_decent</code>	HSL - D
<code>hsl_cent</code>	HSL - MC
<code>hcomm</code>	SC/S - D
<code>advice</code>	SC/A - D
<code>chat_decent</code>	CH/S - D
<code>chat_cent</code>	CH/A - D
<code>chat_advice</code>	CH - MC
<code>game</code>	game being played
<code>period</code>	round
<code>group</code>	three person group within session
<code>original_subject</code>	subject within session
<code>type</code>	0 = Central Manager, 1 = Branch 1, 2 = Branch
<code>message</code>	player's own message
<code>message1</code>	message sent by Branch 1, centralized
<code>message2</code>	message sent by Branch 2, centralized
<code>strategy1</code>	action for Agent 1
<code>strategy2</code>	action for Agent 2

payoff_b1	payoff for Agent 1
payoff_b2	payoff for Agent 2
payoff_manager	manager's payoff, also total surplus
b1chat_row_r1	comm between branches, chat round 1, branch 1
b1chat_row_r2	comm between branches, chat round 2, branch 1
b1chat_row_r3	comm between branches, chat round 3, branch 1
b1chat_col_r1	comm between branches, chat round 1, branch 1
b1chat_col_r2	comm between branches, chat round 2, branch 1
b1chat_col_r3	comm between branches, chat round 3, branch 1
b2chat_row_r1	comm between branches, chat round 1, branch 2
b2chat_row_r2	comm between branches, chat round 2, branch 2
b2chat_row_r3	comm between branches, chat round 3, branch 2
b2chat_col_r1	comm between branches, chat round 1, branch 2
b2chat_col_r2	comm between branches, chat round 2, branch 2
b2chat_col_r3	comm between branches, chat round 3, branch 2
man_suggest_row_g1	advice, suggested row, g1
man_suggest_row_g2	advice, suggested row, g2
man_suggest_row_g3	advice, suggested row, g3
man_suggest_row_g4	advice, suggested row, g4
man_suggest_row_g5	advice, suggested row, g5
man_suggest_col_g1	advice, suggested col, g1
man_suggest_col_g2	advice, suggested col, g2
man_suggest_col_g3	advice, suggested col, g3
man_suggest_col_g4	advice, suggested col, g4
man_suggest_col_g5	advice, suggested col, g5

Variable List, BC Coding Data, Main Experiment

<u>Variable Name</u>	<u>Description</u>
session	session id
true_treatment_ordered	treatments, same order as paper
group	three person group within session
global_group	global id number for groups
period	round
chat_decent	CH/S - D
chat_advice	CH - MC
chat_cent	CH/A - D
game	game being played
strategy1	action for Agent 1
strategy2	action for Agent 2
coordinate	agents coordinate

coordinate_safe	coordinate at safe equilibrium
coordinate_efficient	coordinate at efficient equilibrium
coordinate_other	coordinate at other equilibrium
payoff_b1	payoff for Agent 1
payoff_b2	payoff for Agent 2
payoff_manager	manager's payoff, also total surplus
any_suggestion	Any Suggestion
suggest_safe	Suggest Safe Outcome
suggest_efficient	Suggest Efficient Outcome
last_suggest_efficient	Final suggestion was efficient
agree_proposal	Agreement to Suggestion
discuss_coordinate	Discuss Need to Coordinate
discuss_fairness	Discuss Fairness
discuss_efficient	Discuss Efficiency
discuss_rules	Questions About Rules of the Experiment
discuss_howtoplay	Questions About How to Play
explanation	Explanation
ask_game	Ask What Game Is Being Played (M)
receive_report	report about game being played
truthful	truthfully reveal game
falsehood	lie about game
contradict	Contradict (One tells truth, other lies)
disagreement	Explicit disagreement about game
neither_report	neither agent reported
truth_a1	Agent 1, truthful
truth_a2	Agent 2, truthful
false_a1	Agent 1, lied
false_a2	Agent 2, lied
nummess_man	# messages, manager
nummess_agent	# messages, agents
man_suggest_safe	Manager, Suggest Safe Outcome
man_suggest_efficient	Manager, Suggest Efficient Outcome
man_agree_proposal	Manager, Agreement to Suggestion
man_discuss_coordinate	Manager, Discuss Need to Coordinate
man_discuss_fairness	Manager, Discuss Fairness
man_discuss_efficient	Manager, Discuss Efficiency
man_discuss_rules	Manager, Questions About Rules of the Experiment
man_discuss_howtoplay	Manager, Questions About How to Play
man_explanation	Manager, Explanation

man_ask_game Manager, Ask What Game Is Being Played (M)

Variable List, BC Master Data, Follow Up Experiment

<u>Variable Name</u>	<u>Description</u>
obn	Observation Number (-999 if no message)
time	time elapsed
session	Session
period	Period
late	Periods 10 - 18
message	message text
subject	Subject
group	Group
subjectid	unique subject id
type	Type
messagecode	numeric code for message types
game	Game
strategy1	Agent 1 action
strategy2	Agent 2 action
payoff_1	Agent 1 payoff
payoff_2	Agent 2 payoff
payoff_manager	Manager payoff
surplus_babbling	surplus from babbling equilibrium

List of tables and programs

The provided code reproduces:

- ☒ All numbers provided in text in the paper
- ☒ All tables and figures in the paper

Note: To be precise, it generates the numbers used to generate the tables and figures. These were then copied into MS Word or MS Excel to create the tables and figures.

- ☐ Selected tables and figures in the paper, as explained and justified below.

References

Acknowledgements

Some content on this page was copied from [Hindawi](#). Other content was adapted from [Fort \(2016\)](#), Supplementary data, with the author's permission.